D.Y.PATIL COLLEGE OF ENGINEERING

Akurdi, Pune -411044

LABORATORY JOURNAL INDEX

COURSE: LABORATORY PRACTICE III

(BE COMPUTER)

Academic Year 2025-26 Sem-I

INDEX

Name of Student: Digvijay Deelip Mangaonkar

Class: B.E **Batch:** A **Branch: Computer Engineering**

Course with Code: Laboratory Practice III (410246)
Companion Course: Design and Analysis of Algorithms (410241), Machine Learning(410242), Blockchain Technology(410243)

Sr. No.	Title of Assignments	Date of Performance	Sign/ Remarks
	Group A: Design and Analysis of Algorithms		
1	Write a program non-recursive and recursive program to calculate Fibonacci numbers and analyze their time and space complexity.	3/7/25	
2	Write a program to implement Huffman Encoding using a greedy strategy.	4/7/25	
3	Write a program to solve a fractional Knapsack problem using a greedy method.	10/7/25	
4	Write a program to solve a 0-1 Knapsack problem using dynamic programming or branch and bound strategy.	17/7/25	
5	Design n-Queens matrix having first Queen placed. Use backtracking to place remaining Queens to generate the final n-queen's matrix.	24/7/25	
6	Mini Project-I	1/8/25	
	Group B: Machine Learning		
Sr. No.	**Title of Assignments**	**Date of Performance**	**Sign/ Remarks**
7	Predict the price of the Uber ride from a given pickup point to the agreed drop-off location. Perform following tasks: 1. Pre-process the dataset. 2. Identify outliers. 3. Check the correlation. 4. Implement linear regression and random forest regression models. Evaluate the models and compare their respective scores like R2, RMSE, etc.	8/8/25	
8	Classify the email using the binary classification method. Email Spam detection has two states: a) Normal State – Not Spam, b) Abnormal State – Spam. Use K-Nearest Neighbors	14/8/25	

	and Support Vector Machine for classification. Analyze their performance.		
9	Given a bank customer, build a neural network-based classifier that can determine whether they will leave or not in the next 6 months. Dataset Description: The case study is from an open-source dataset from Kaggle. The dataset contains 10,000 sample points with 14 distinct features such as CustomerId, CreditScore, Geography, Gender, Age, Tenure, Balance, etc.	21/8/25	
10	Implement K-Nearest Neighbors algorithm on diabetes.csv dataset. Compute confusion matrix, accuracy, error rate, precision and recall on the given dataset	28/8/25	
11	Implement K-Means clustering/ hierarchical clustering on sales_data_sample.csv dataset. Determine the number of clusters using the elbow method.	4/9/25	
12	Mini Project-II	12/9/25	

Group C: Blockchain Technology

Sr. No.	Title of Assignments	Date of Performance	Sign/ Remarks
13	Installation of Metamask and study spending Ether per transaction.	19/9/25	
14	Create your own wallet using Metamask for crypto transactions.	26/9/25	
15	Write a smart contract on a test network, for Bank account of a customer for following operations: • Deposit money • Withdraw Money • Show balance	3/10/25	
16	Write a program in solidity to create Student data. Use the following constructs: • Structures • Arrays • Fallback Deploy this as smart contract on Ethereum and Observe the transaction fee and Gas values.	10/10/25	
17	Write a survey report on types of Blockchains and its real time use cases.	17/10/25	
18	Mini Project-III	30/10/25	

Signature of Student

Course Coordinator

Assignment - 1

Page No.
Date

Title :
Fibonacci Numbers

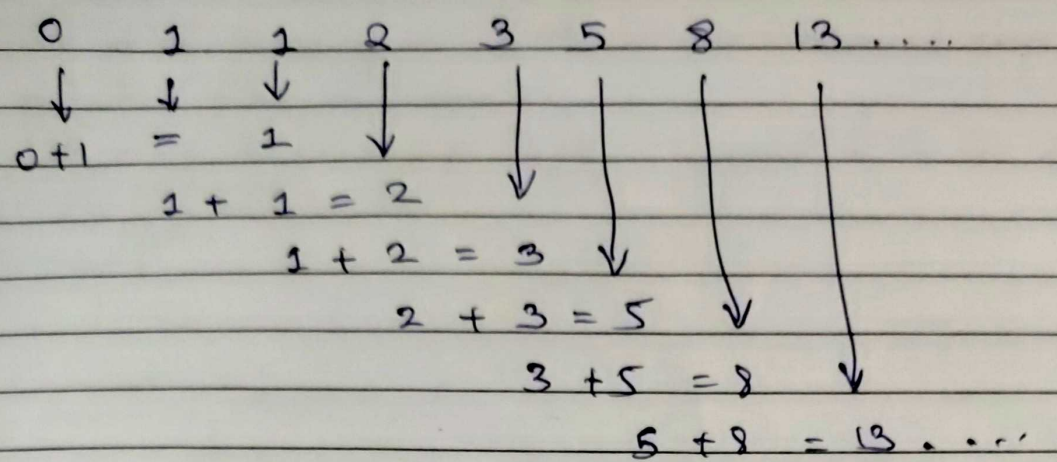
Objectives :
To learn how to count steps & henceforth time complexity of algorithm.
Learn how to implement algorithms that follow algorithm design strategies like Divide N Conquer, Greedy, Dynamic Programming, Backtracking, Branch N Bound, etc.

→ **Problem Statement :**
Write a program to Calculate fibonnacci Number & find it step count.

→ **Software & Hardware Requirement :**
64 bit open source Linux or its derivative, Python 3.9 or higher.

Hardware
- Intel i3 or higher
- 512 GB SSD, 4 GB RAM

→ **Theory :**
The fibonacci series is a sequence where each number is sum of previous 2 numbers of sequence.
The 1st two numbers of fibo series are 0 & 1 and are used to generate the fibo series



In mathematical terms, the number of nth posith can be represented by

$$\boxed{F_n = f_{n-1} + F_{n-2}}$$

The 1st ten numbers in fibo series are:

0, 1, 1, 2, 3, 8, 13, 21, 94

→ Complexity Analysis:

- Time Complexity: O(n)
- Space Complexity: O(n)

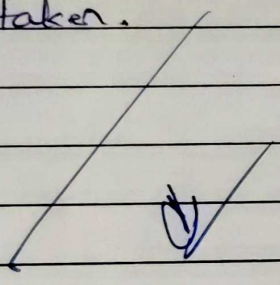
→ Algorithms:

1. Start
2. Initialize variable 'step to 0'.
3. Read value 'n'
4. Initialize two variables f1 & f2 to 0 & 1
5. Initialize empty list 'fib'

6. If n if 2, append f_1 & f_2 & increment step by 2 print 'fib' & 'steps'.
7. If n is 3 or more firstly append f_1 & f_2 to fib.
$fib_{next} = f_1 + f_2$
8. STOP

→ Condution:

Thus, we successfully calculated fibo numbers upto the number inputted by user & also number of steps taken.



Assignment - 2

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Date

→ Title :
Job sequencing with deadlines.

→ Objectives :
To understand the working of Job sequencing in computer under deadlines.

→ Problem Statement :

Implement Job sequencing with deadlines using a Greedy method.

→ Requirements :

- Software : 64 bit Linux
  Python 3.9 or higher
  Hardware : Processor → Intel i3
  512 GB SSD 4GB RAM.

→ Theory :

Greedy Algorithm :

Greedy Algorithm are a class of algorithm that make locally optimal choices of each global step with the hope of finding a global optimum soln.

Examples of Greedy Algorithm :
- Fractional Knapsack.
- Dijkstra Algorithm

iii) Kruskal Algorithm
iv) Huffman Coding

- Application of Greedy Method in DAA include reducing data size by removing redundant information dividing images into regions with similar information etc.

- Job Scheduling:
Given a number of jobs with a starting time & ending time, they need to be scheduled in such a way that maximum profit is received with in the maximum deadline

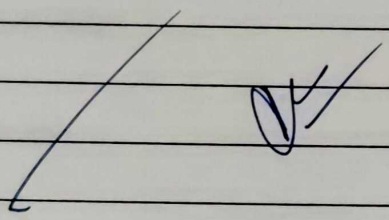
- Algorithm:

1. find the maximum deadline value from input set of jobs.
2. Once the deadline is declared, arrange the jobs in ascending order of their profits
3. Select the jobs with highest profits, their the period not exceeding the maximum deadline
4. Selected set of jobs are output

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→ Conclusion :

Thus, we successfully implemented job sequencing with deadline using a Greedy approach.



Assignment - 3

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Date

→ Title :

Knapsack Problem

→ Objectives :

To learn how to implement Knapsack Problem & understand its importance.

→ Problem Statement :

- Write a program to solve fractional knapsack using a Greedy Method.
- Write a program to solve 0-1 knapsack using DP or Branch N Bound.

→ Requirements :

Software : 64 bit Ubuntu Linux.
Python 3.9 or higher.
Hardware : Intel i3
512 GB SSD 4GB RAM

Knapsack Problem :

The Knapsack Problem is an example of the combitional optimizatn problem.

- Types of Knapsack Problem :

1. fractional Knapsack.
2. 0/1 Knapsack
3. Bounded knapsack.
4. Unbounded Knapsack.

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- Fractional Knapsack:

In fractional knapsack we can break the items for maximizing the total value of knapsack.

Objects	Profits	Weights	(P/w) Ratio.
1	5	1	5
2	10	3	3.33
3	15	5	3
4	11	4	2.7
5	8	1	8
6	9	3	3
7	4	2	2

Weight of knapsack (W_K) = 15.

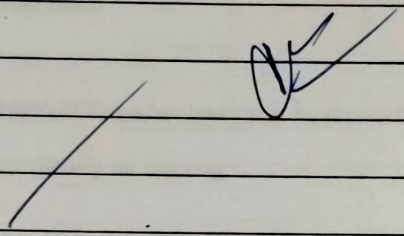
- Algorithms:

1. Start
2. Calculate profit in weight ratio for each item.
3. Sort the item in descending order based on this ratio
4. Initialize total value to 0.
5. If entire item is added, add the update value.
   If only a fract^n of the item can be added, add the fract^n & update value.
6. STOP.

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→ Conclusion :

Thus, we have successfully studied greedy strategies & used them to solve fractional knapsack problems



Assignment - 5

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→ Title :
8-Queen Matrix

→ Objectives :

To understand the concept of a popular puzzle, the N-Queens

→ Problem Statement :

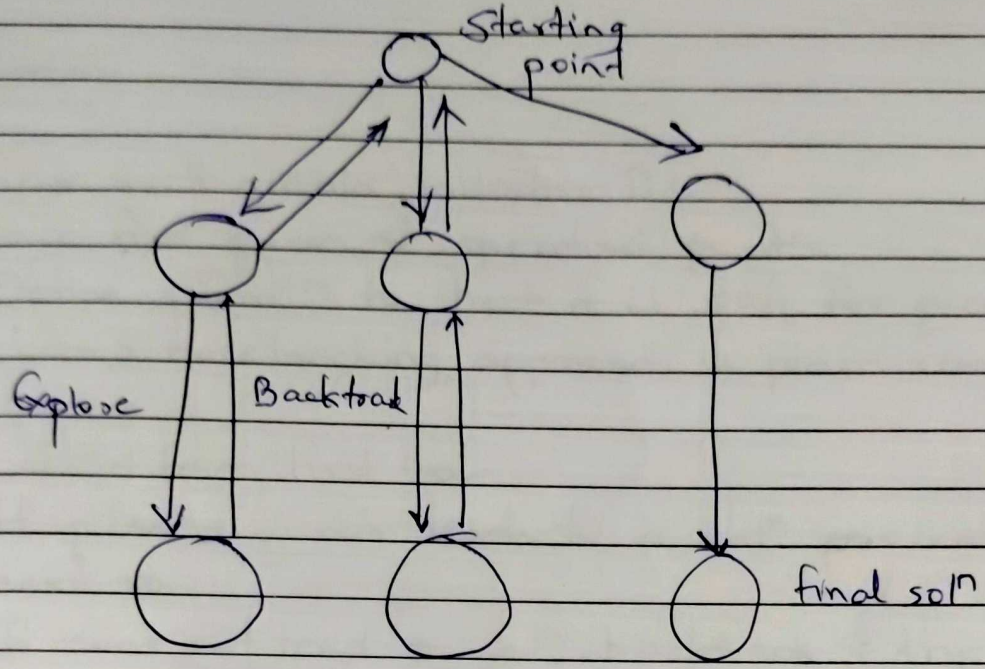
Design 8-Queens Matrix having first Queen placed use backtracking to place remaining queens to get final 8-queens matrix.

→ Software & Hardware Requirement :

- Software : 64 bit Ubuntu linux
  Python 3.9 or higher
- Hardware : Intel i3
  512 GB SSD 4 GB RAM

→ Theory :

Backtracking is a problem solving algorithm technique that involves finding a soln incrementally by trying different options & unclosing them if they lead to a dead end.



- **N-Queens:**

"N-Queens is the problem of placing N-chess queens on a N×N chess board so that no two queens attack each other.

Q	x	x	x	x	x	x	x
x	x	x	x	x	x	Q	x
x	x	x	x	Q	x	x	x
x	x	x	x	x	x	x	Q
x	Q	x	x	x	x	x	x
x	x	x	Q	x	x	x	x
x	x	x	x	x	Q	x	x
x	x	Q	x	x	x	x	x

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- Algorithm:

1. Create 8×8 matrix, initialize DS.
2. Place first queen at specified posith.
3. Define a funct^n to check if is safe for queen.
4. Use a backtracking approach to place remaining queens
5. Start from first row.
6. If placing queen leads to a sol^n proceeds to the next row.
7. If does not lead to sol^n, backtrack & try next sol^n.
8. STOP

- Time Complexity:
  $O(n^2)$ where 'n' is no. of queens

- Space Complexity:
  $O(n^2)$.

→ Conclusion:

Thus, we successfully studied & implemented 8-queen problem using backtracking

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→ Title :

Program to Generate Binomial Coefficients using Dynamic Programming.

→ objective & Problem statement :

To write a program that computes binomial coefficient efficiently using the concept of dynamic programming.

→ Software & Hardware Requirements :

- Software : C/C++ Compiler (eg. GCC, Turbo C++).
- Hardware : Intel i3, 512 GB SSD, 4GB RAM

→ Theory :

Binomial coefficients are the coefficients in the expansion of $(a+b)^n$.
They are represented as $C(n,k)$ or ${}^nC_k$ & defined as

$$C(n,k) = \frac{n!}{k!(n-k)!}$$

Using Dynamic Programming, we avoid redundant calculations by storing intermediate results in a 2D array where :

$$C(n,k) = C(n-1,k+1) + C(n-1,k).$$

with base cases:

$$C(n,0) = C(n,n) = 1$$

→ Algorithm:

1. Start
2. Input value of n (number of rows).
3. Create a 2D array C[n+1][n+1]
4. for each i from 0 to n:
   a. for each j from 0 to i:
      - If j=0 or j=i, set C[i][j] = 1
      - Else set C[i][j] = C[i-1][j-1] + C[i-1][j]
5. Print the binomial coefficients in tabular form.
6. Stop

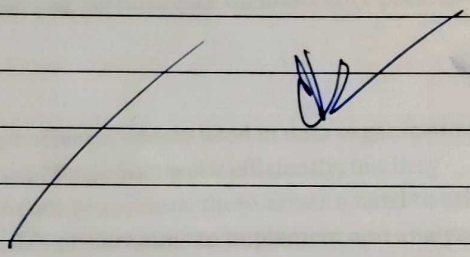
→ Code:

```
#include <stdio.h>
void printBinomialCoeff(int n) {
    int C[n+1][n+1];
    for (int i=0; i<=n; i++) {
        for (int j=0; j<=i; j++) {
            if (j == 0 || j == i) {
                C[i][j] = 1;
            else
                C[i][j] = C[i-1][j-1] + C[i-1][j];
            printf(C[i][j]);
    }
    printf("\n");
}
```

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→ Conclusion :

The program successfully generates binomial coefficients using dynamic programming, reducing redundant computations & optimizing time complexity compared to the recursive approach.



Assignment -7

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Date

→ Title :

Uber Price Prediction :- Linear Regression & Random forest.

→ Objectives :

Learn effect of data preprocessing on the performance of the ML algorithms.

→ Problem Statement :

Predict the price of Uber Ride from a given point to agreed drop point.

Perform following topics :

1. Preprocess the dataset
2. Identify Outliers
3. Check the correlations

→ Requirements :

- Software : 64-bit Linux.
- Hardware : Intel i3, 512 GB HDD 4 GB RAM

→ Theory :

- Linear Regression :

Linear Regression is a basic statement supervised

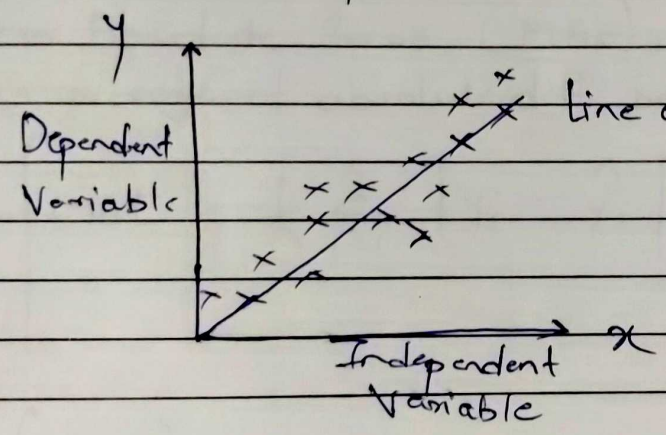
learning algorithm used for predicting a continously target variable based on relationship beth input feature & output

The main goal is to find the line that best fits the data

fqn:

$$y = \beta_0 + \beta_1 x_1 + c$$

- y → target variable
- β_0 → Intercept
- β_1 → slope
- x_1 → Input term.
- c → Random error



II. Random forest :

Random forest is an assemble learning algorithm mainly used for classificat^n & regression tasks

It combines multiple decision trees to improve the predective performance & reduce overfitting

The final prediction is based on the majority voting in classificat^n

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III. Evaluation Matrix

- Mean Squared Error (MSE).

It is the average of square difference bet^n actual & predicted value.

$$MSE = \frac{1}{n} * \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

y_i = actual ; $\hat{y}_i$ = predicted

- Mean Absolute Error (MAE).

It is average of absolute diff bet^n actual & predicted valu

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

- R^2 (Error).

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

$\bar{y}$ = mean of actual val

⇒ Conclusion :

Thus, we successfully implemented & compared the linear regression & random forest models.

Assignment - 8

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→ Title:

Email Classificatn - KNN, SVM.

→ Objectives:

Learn effect of data preprocessing on the performance of ML algorithm.

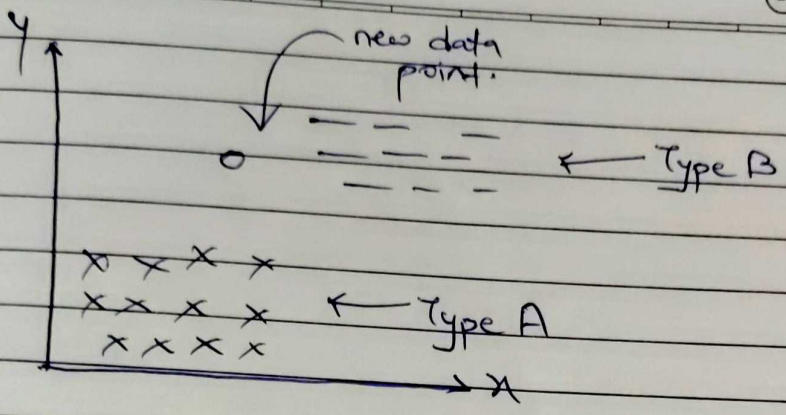
→ Problem statement:

- Classify the email using binary classificatn method. Email Spam detection has two states:
  a) Normal state - Not SPAM
  b) Abnormal state - SPAM.

- Use K-nearest neighbours & SVM for classification.
- Analyze the performance.
- Use email CSV dataset from Kaggle.

→ Theory:

- KNN (K- Nearest neighbour).
- KNN is a simple, supervised learning algorithm used for classification.
- It finds the K-closet data-points to a given input & making predictions based on majority class.



SVM (Support Vector Machine).

SVM is a powerful SLA used for classificatn. It aims to find the optimal boundary that best separats data point of different classes in a feature space.

II. Accuracy measures based on the values of confusion matrix are

- Confusion Matrix

Actual \ Predicted	Positive	Negative
Positive	TP	FN
Negative	FP	TN

- Accuracy $= \dfrac{TN + TP}{TN + TP + FN + FP}$

- Precision $= \left[P = \dfrac{TP}{TP + FP} \right]$

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- **Recall**

$$\left[\text{Recall} = \frac{TP}{FP + FN} \right]$$

- **Specificity**

$$\left[S = \frac{TN}{TN + FP} \right]$$

→ **Conclusion:**

Thus, we successfully learnt, implemented compared classification, using KNN & SVM.

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→ Neuron

- An estimated 10^{11} (10 Billion) nerve cells as neurons make up the human brain.
- Electrical Signals, which are brief impulsed or spikes in the voltage of cell wall or membrane are used by neurons to communicate

I. Steps in building a Neural Network based classificatn:

- Data Preparatn:

i) Collect & process the data to ensure consistent scaling

ii) Define the model architecture

iii) Hidden layer: Define no of layers & neurons in each layer based on data complexity.

iii) Output layer: Use 1 neuron with sigmoid activatn funct^n for binary classificatn.

- Compile the Model:

i) Loss funct^n: Use categorical cross entropy for multi-class & binary cross entropy.

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4. Train the Model:

i) Use Backpropagatn an optimizer to adjust the weights by minimizing loss functn.

ii) Use validatn data to monitor model performance & avoid overfitting

5. Evaluate the Model:

After training, test the model on unseen test data to evaluate it performance

→ Conclusion :

Thus, we have successfully studied a neural network based classifier.

Assignment - 10

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Date

→ Title :

Diabities Dataset → KNN

→ Objective :

Learn effect of data preprocessing on the performance of ML algorithm.

→ Problem statement :

Implement K-Nearest neighbours an diabities dataset Compute confusion matrix, accuracy, error rate precision & recall on the given dataset

→ Requirements :

- Software : 64 bit Linux, Python 3.9
- Hardware : 512 GB RAM, Intel i3, 4GB RAM.

→ Theory :

KNN (k-Nearest Neighbours).
KNN is a simple supervised learning algorithm used for classificatn task.

- How does the KNN work ?

1. Select the number of neighbours (k) to consider
2. Calculate the distance betn the test-point & all other points.

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3. Identify the K-points that are closed to the test points
4. Count the number of data points belonging to the different class
5. The class that appears most frequently among the K-neighbours

- Advantage :

1. Simple to understand & implement.
2. It makes no assumption about the underlying data distribution.

- Disadvantages :

1. It is computationally expensive to measure distance.
2. It is sensitive to irrelevant features.
3. Performance depends on K, if K↓ overfitting, K↑ underfitting.

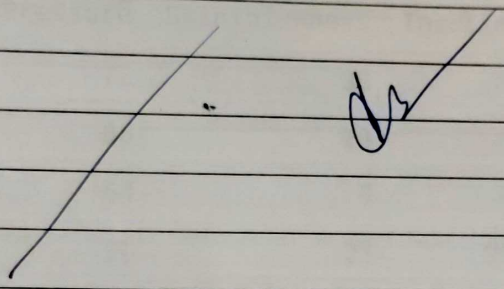
→ Applications :

1. Image Recognition, classifying images
2. Recommendation system based on user preferences
3. Medical diagnosis, classifying diseases.
4. Pattern recognization, speech recognization, gesture recognization.

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→ Conclusion:

Thus, we successfully studied & implemented KNN algorithm on the diabities data set & tested it on various performance metrics



Assignment - 11

→ **Title:**

Implementation of K-Means / Hierarchical Clustering on sales_data_sample.csv & Determination of Optimal Clusters using Elbow Method.

→ **Objective:**

To apply K-Means & Hierarchical clustering algorithms on the sales dataset & determine the optimal number of clusters using the Elbow method.

→ **Software & Hardware Requirements:**

- Software: Python 3, Jupyter Notebook
- Hardware: Intel i3, 512 GB SSD, 4 GB RAM

→ **Theory:**

Clustering is an unsupervised learning technique that groups data points based on similarity.

- K-Means Clustering:
  Divides data into K clusters by minimizing the sum of squared distances bet^n points & their clusters centroids

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o Hierarchical Clustering:

- Builds a hierarchy of clusters either th^ agglomerative or divisive approach.
- The Elbow method helps determine the optimal number of clusters by plotting the Within-Cluster-Sum-of-Squares (WCSS) against different values of k & identifying the point where the rate of decrease sharply changes.

→ Algorithm:

1. Start
2. Load & preprocess sales_data_sample.csv.
3. Select relevant numerical features for clustering
4. Standardize the data
5. Apply K-Means for k=1 to 10 & compute WCSS
6. Plot WCSS vs. k to find 'elbow',
7. Re-run K-Means with optimal k value
8. Visualize & analyze clusto results
9. Stop

→ Conclusion:

The Elbow method successfully identified the optimal number of clusters for the sales_data_sample.csv dataset.

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K-Means effectively grouped similar sales record helping to uncover customer or sales behaviour patterns for business insights

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Assignment - 13

→ Title:
Installation of Metamask

→ Objective:
- Learn how to install metamask on a firefox browser
- Understand & explore the working of Blockchain Technology & its applications.

→ Problem Statements:
Installatn of Metamask & study spending per transactn.

→ Software & Hardware Requirements:

- Software:
  64 bit Open Source Ubuntu.
- Hardware:
  Intel i3, 512 GB SSD, 4 GB RAM.

→ Theory:

#Metamask:
It is a popular cryptocurrency wallet & browser extension that allows users to manage, store & interact with Ethereum & other.
ERC → 20 tokens
with metamask users can securly store private keys & perform crypto transactn directly from their web browser.

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→ Steps to install Metamask on Ubuntu (firefox) :

1. Open firefox on your Ubuntu system.
2. Go to the official metamask website on directly visit the firefox add-one store.
3. On the metamask page, click. 'Add to firefox'.
4. A popup will ask for permission. Click 'Add' to confirm.
5. After installation, Metamask will appears as a extension in the brocoses toolbar.
6. Optionally, you can point the extension by clicking the extension icon in the toolbars & selecting 'pin'

→ Steps for spending Ether :

1. Initiate a transactn

Open a Metamask & enter reciptents address & the amount of ether you want to send

2. Check the fees :

3. Confirm the transactn :
Click on 'Confirm' to send the ETH

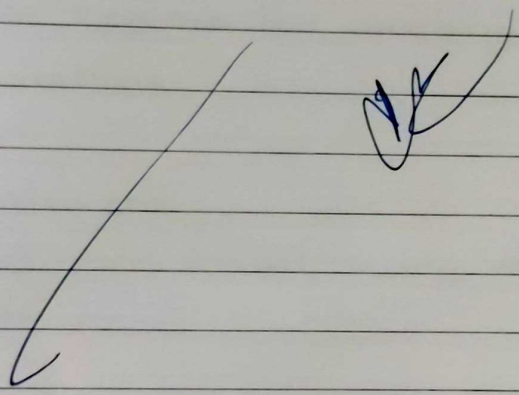
4. Transactn status :

You can track the transactn status on ethers scan using hash

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→ Conclusion!

Thus, we have successfully installed Metamask & studied spending ether per transactions.



FOR EDUCATIONAL USE

Assignment - 14

→ Title:
Metamask Wallet Creation

→ Objective:
- Learn how to create a coallet using metamask for cryptocurrency in Blockchain.
- Understand & explore the coorking of Blockchain Technology & its applications.

→ Problem Statement:
Create your ocon coallet using metamask for crypto transaction.

→ Softcoare & Hardcoare Requirements:

- Softcoare : 64 bit Open source Ubuntu.
- Hardcoare: Intel i3, 512 GB SSD, 4 GB RAM

→ Theory:

Cryptocoallet:

* It is nothing but softcoare coding used to store, send & receive cryptocurrency.
- It is a place that stores public & private keys safely & allocas you to interact coith various blockchain.

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→ How to create a wallet on Metamask ?

1. Click on the metamask extension icon in your browser, click on 'Get started'.
2. Choose 'Create a Wallet'.
3. Create a strong password for accessing your coalled on your local browser. Agree to the terms and click on 'Create'.
4. Metamask coill generate at 12-word secret recovery phrase. Store it at a safe place.
5. Once the phrase is confirmed you'll be directed to your neco wallet.

→ Types of Crypto Wallet :-

I. Software Wallet :

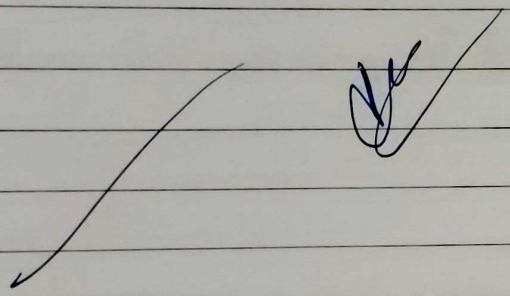
1. Desktop Wallet :
It is a wallet which is doconloaded & uninstaled on desktop. It can only be accessed in the system in which it is installed.
2. Mobile Wallet :
It is an app which can be used anytime anywhere
3. Online Wallet.

II. Hardware Wallet :
They store crypto offline. Public / private keys are stored in USB's.

FOR EDUCATIONAL USE

→ Conclusion:

Thus, we successfully created our own wallet on metamask for crypto transactions.



FOR EDUCATIONAL USE

Assignment - 15

→ Title:

Smart Contract for Bank Account:

→ Objectives:

- Learn how to write a smart contract for cryptocurrency in Blockchain.
- Understand & explore the working of Blockchain Technology & its applications

→ Problem Statement:

Write a smart contract on a test network for Bank account of a customer for following operations

- Deposit Money
- Withdraw Money
- Show Balance.

→ Software & Hardware Requirements:

- Software: 64bit open source Ubuntu.
- Hardware: Intel i3, 512 GB SSD, 4 GB RAM.

→ Theory:

#Smart Contract:

- A smart contract is a self-executing contract with the terms of agreement written into code
- It runs on a ~~blockfire~~ blockchain & automatically enforces & executes the terms of the contract when predefined conditions are met, without the need of intermediary

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→ Types of Smart Contracts:

1. Smart Legal Contracts:
These are legally enfocable & require the partices fullfill their responsibilities. failure to do so can result in strict legal action.

2. Application Logic Contract:
These comprise of application-based code that left overs in-sync coith other contracts. It enables communication beth altred devices, like IoTA coith blockchain.

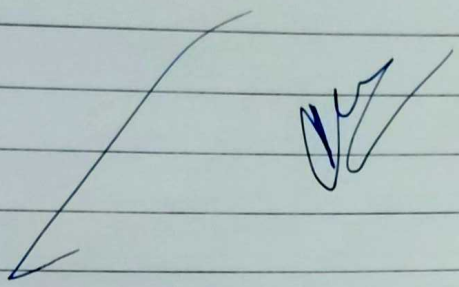
3. Decentralized Autonomous Organizations:
These are blockchain groups that are guranteed to precise rules coded into contracts mutual coith governance mechanism. Henceforth any actions taken by the community members get reploaded by a self enforaing code.

→ Solidity:
- It is a high-level, statically typed programming contracts on Ethereum Blockchain.
- It is similar to javascript, c++ & python in syntax & is specifically designed to develop decentralized that can run on EVM.

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Conclusion:

Thus, we successfully studied & implemented smart contracts for bank account of a customer



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Assignment - 16

→ Title:

Solidity program for student data.

→ Objectives:

- Learn how to write a program in Solidity & deploy it as a smart contract.
- Understand & explore the working of Blockchain technology & its applicatn.

→ Problem Statement:

write a program in Solidity in create student data.
Use the following constructs.

- Structures
- Arrays
- Fallback

→ Deploys this as smart contract on Ethereum & observe the transaction value.

→ Software & Hardware Requirements:

- Software: 64 bit Ubuntu
- Hardware: Intel i3, 512 GB RAM, 4GB RAM

→ Theory:

Solidity is a high level, statically-typed programming language, used to write.

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→ Data Types in Solidity :

1. Boolean :
Represent true or false values.
e.g. bool is complete = false ;

2. Integer :
i) Signed Integers (int) : Positive or Negative . The size can be specified (e.g. int 8 , int256).
ii) Unsigned Integers : Only non-negative numbers
e.g. uint256 , total supply = 1000 ;

3. Address :
Holds a 20 byte Ethereum address , typically used to store account or contract addresses.
e.g. Address = 0x123456789dcb def12456 78 Dcdefgh

4. String :
A sequence of characters , they are immutable
e.g. String public name = 'ABC' ;

5. Bytes :
i) fixed : size (bytes1 to bytes32) : Array of bytes of fixed length eg. bytes32 fix data = 'Hello' ;

ii) Dynamic Size : Adynamic array of bytes

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→ A Basic Solidity Program:

Given below is a simple smart contract to store & retrieve a number.

e.g. Solidity ^0.8.0;

```
Contract simplestorage {
        unit public storedData;
        function set (unit x) public {
                storedData = x;
        }
        function get () public view returns (unit) {
                return storedData;
        }
}
```

→ How to deploy smart contract on Ethereum:

1. Install Metamask:

2. Get test Ether.
3. Install Remix IDE.
4. Write & compile the contract.
5. Deploy the contract.
6. Interact with contract.

→ Conclusion:

Thus, we successfully studied solidity & work a program to create student data successfully.

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Assignment - 17

Page No.
Date

→ Title:
Survey Report on types of Blockchains & their real-time use case.

→ Objective:
To study different types of blockchain architecture & explore their real-world applications across various domains.

→ Software & Hardware Requirements:

- Software: MS word, Google docs.
- Hardware: Intel i3, 512GB SSD, 4GB RAM

→ Theory:

Blockchain is a distributed ledger technology that records transaction securely, transparently & immutably. It operates without a central authority, enabling trust & verification thr consensus mechanisms like Proof of Work or Proof of Stake.

There are mainly 4 types of Blockchains:

1. Public Blockchain
   Open to everyone; anyone can participate & validate transactions.

2. Private Blockchain
   - Controlled by a single organization
     e.g. Hyperledger fabric, Coorda.

3. Consortium Blockchain:
- Managed by a group of organizations; offers shared control & transparency.
- Ex. Energy Web Foundation, R3 Corda.

4. Hybrid Blockchain:
- Combines features of public & private blockchains for flexibility.
- Ex. IBM Food Trust, Dragonchain.

→ Real-Time Use Cases:

- Finance: Cross-border payments & smart contract (Ripple, Ethereum).
- Supply chain: Product tracking & authentication (IBM Food Trust).
- Healthcare: Secure sharing of patient records (MedRec).
- Government:

→ Conclusion:
Different types of blockchains serve specific use case based on transparency, control, & scalability needs.